Abstract

Egypt faces several grand challenges over the years, like the lack of alternative energy and improving the sciences and technologies. The purpose of the study is to reduce the emissions of fossil fuels.

Thus, some cities suffer from this problem, for instance, Helwan, due to the expansion of industrial areas and non-renewable power stations, which endanger Helwan's health and environment through air pollution. To address this issue, a Project of an O-wind turbine will be developed to capture wind from all directions (omnidirectionally) and generate electricity. Additionally, it uses eco-friendly materials, with at least one material that should be recycled, therefore more sustainable and eco-friendlier for the environment. The materials include steel, aluminum, and concrete. The turbine design will be based on a tetrahedron forming a spherical structure, composed of channels with a larger input than output. When a turbine passes wind through channels, the pressure difference between the input and output causes it to generate energy. Furthermore, the project must meet specific design requirements. They involve generating at least **150** joules within **5** minutes, with at least a **10**% performance improvement using a gearbox. Prototype volume must not exceed **0.25** m^3. The used materials are eco-friendly, and at least one must

be recycled. Prototype doesn't rely on solar energy, without toxics, explosives, or flammables.

Introduction

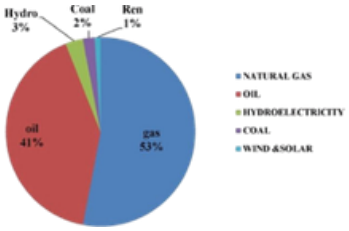

Figure 1/ Energy consumption in Egypt

Egypt faces interconnected grand challenges that impede its development, including waste management, which is the process of recycling and safe disposal of waste materials, contributing to decreased air, water, and soil pollution. Moreover, using alternative energy can reduce pollution caused by reliance on non-renewable sources, including fossil fuels, which account for approximately 90% of Egypt's energy. **As shown in Figure 1**, the problem to be solved is the pollution caused by reliance on non-renewable energy, which offers many harmful effects for Egypt. Unfortunately, ignoring this issue significantly harms public health. Air pollution contributes to approximately 23,200 deaths in Egypt; the environment is also severely impacted. Egypt spent 26 billion dollars on fossil fuels in 2012. Therefore, Egypt ranks as the eighth-highest spender on fossil fuels worldwide; furthermore, reducing dependence on fossil fuels will decrease pollution in Egypt. Currently, 76.5% of electricity generation comes from natural gas, 13.7% from oil, and only 3.3% from renewables. For instance, Helwan city hosts one of Egypt's largest fossil-fuel power plants, the Helwan South Power

Station. It generates power equivalent to about 90% of the Aswan High Dam. According to the World Health Organization (WHO), PM2.5 levels in Helwan are about seven times higher than the safe limit.

Solving this problem requires analyzing previous projects. Firstly, the Ras Ghareb Wind Farm serves as a key example. It uses advanced technology with minimal energy loss, achieving high efficiency, with a capacity of 262.5 MW, powering approximately 500,000 Egyptian households; however, this technology is imported, increasing costs and reliance on loans and investors. Secondly, the Alta Wind Energy Center (AWEC), a wind energy project in the United States, is characterized by sustainability and eco-friendliness, providing 100% clean energy. On the other hand, it requires a large area due to the vast spaces between each turbine, with noise and visual impact.

Consequently, the selected solution is an O–wind turbine, designed in spherical geometry, with a tetrahedron shape. to capture wind from vertical, horizontal, and diagonal directions and convert it into rotational motion on a unique axis, which is converted into electrical energy. This approach consists of channels having entrances larger than the existing ones in variable proportions. Thus, when wind passes through the channels, the difference between the entrance and exit creates a pressure difference that generates electricity. Additionally, the prototype dimension is 0.128m³. Ultimately, it must meet the design requirements, including

using eco-friendly and recycled materials with low environmental impact and high energy efficiency, achieving at least 150 joules of energy, and a minimum 10% improvement in efficiency. The dimension doesn't exceed 0.25m³.

Materials and Methods

Name	Quantity	Figure	Description and Usage
Polyvinyl Chloride (PVC) sheet	0.7 M		It is known for its high strength-to-density ratio and chemical resistance. It is used in making the inner tetrahedron.
HDPE (High-Density Polyethene) sheet	0.9 M		HDPE is a flexible, waterproof, and lightweight plastic. It is used in making the outer curve of the prototype.
Corrugated plastic board	$1M$		The corrugated plastic board is lightweight and air-resistant. It is used in making the inner blades.
Reclaimed wood	1.6 M		Reclaimed wood is durable and eco-friendly. It used to be in the basement of the prototype.

Recycled iron pipe	0.8 M		Recycled iron pipe is used in Construction
Medium-Density Fiberboard (MDF)	1 board		It is an engineered wood product. It is used for making the basement.
Roller Bearing	2 (12mm)		It is a precision component that reduces friction between moving parts. It is used to make the turbine while revolving have low friction.
DC Motor 24V	1		A 24V DC motor is an electric motor that runs on 24 volts by converting mechanical energy into electrical energy. It is put under the prototype.
Acrylic sheet	1		It is a strong, lightweight material. It is used in making the gears.
MDF spray	2		It is a sticky plastic spray that is used to stick the prototype plastic.

Methods:

1) Before constructing the prototype, the 3D design was made by SolidWorks, **as shown in Figure 2,** to determine the prototype's shape.

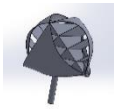

Figure 2/ 3D-design of the prototype

2) The PVC plastic was cut into 4 triangles to make the inner tetrahedron shape with a side length of 32.65 ± 0.05, **as shown in Figure 3.**

Figure 3/ The inner triangle shape

3) After cutting the PVC sheets, the HDPE was cut into 4 triangular shapes with sides 36 * 36 * 44 to make the input double the output to make the turbine rotate, **as shown in Figure 4.**

Figure 4/ The outer triangle shape

4) The HDPE triangle curve was adhered to the base PVC triangle using MDF spray, **as shown in Figure 5.**

Figure 5/ The structure of the turbine's face

5) Then the inner blades were made of corrugated plastic and were stuck to the adhered part by the MDF spray, **as shown in Figure 6.**

Figure 6/ The inner blades

6) The 4 adhered parts were stuck together to make the turbine's spherical shape, **as shown in Figure 7.**

Figure 7/ the upper turbine

7) After that, the basement was made of wood, and the upper turbine was put on it, **as shown in Figure 8.**

Figure 8/ the turbine with the basement

8) Under the shaft, the DC motor was put to generate electricity.

9) After that, the gearbox was made with a ratio of 1:5

10) Then the circuit was made to make an LED light

11)The prototype was enhanced to achieve the 10% efficiency by putting outer blades of eco-friendly material HDPE, **as shown in Figure 9**.

Figure 9/ the enhancement

Test plan

To ensure the efficiency of the prototype in producing about 150 joules within five minutes. Moreover, to evaluate an enhancement of about 10% in performance. The prototype was tested to ensure its stability and efficiency according to the design requirements.

1. The prototype was placed in the school field to capture natural wind conditions, where the wind velocity was approximately 5 m/s.
2. The multimeter was used to measure the voltage and current. Moreover, it was used to calculate the energy using the formula $E = I \times V \times T$,where time is 5 minutes
3. The prototype was oriented toward the wind and started rotating to achieve the required output.
4. To meet the design requirements, including enhancements, the prototype was designed to concentrate wind energy on the spherical structure to contribute to an increase in output efficiency of about 10%.

Results

Negative result

In constructing the prototype, PVC plastic was used to make the tetrahedron shape that is located inside the turbine. The PVC plastic in the tetrahedron was sticked used the silicon wax. The negative result is that the wax doesn't have the ability to stick the PVC plastic to itself. In addition, while constructing the turbine, the shaft was fairly heavy. That made the turbine struggle to rotate, which reduced the RPM, making the turbine generate less electricity. The problem was solved by adding additional wide blades on the outlets to increase the surface area facing the airflow. making the turbines' outlets capture more air to revolve. That made the turbine revolve faster despite the heavy weight of the shaft.

Positive result

In the test plan, there were positive results obtained, which made the prototype achieve the design requirements. In testing the prototype using a digital multimeter, the prototype has successfully achieved **153** joules in **5** minutes by producing an average voltage of **2.7** volts and an average current of **0.189** amperes, **as shown in Figure 10**.

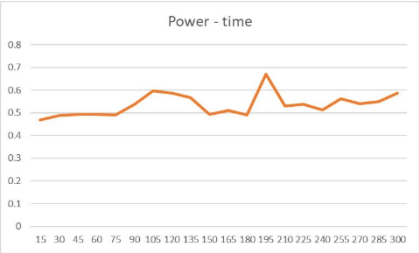

Figure 10: power before enhancement

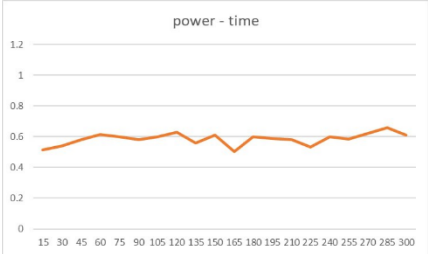

Figure 11: power after enhancement

The prototype has successfully achieved **10%** improvement in efficiency by producing **170** joules in **5** minutes with an average voltage of **2.9** volts and an average current of **0.195** amperes, **as shown in Figure 11.**

Analysis

Torque

To know the force that the turbine needs to revolve, the rotational mechanics were applied to use the torque concept, **as shown in Figure 12**. Torque is used to measure the required force on an object to make it rotate around an axis called the pivot point. The torque is calculated using **Equation 1**.

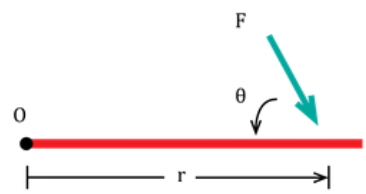

Figure 12: the torque principle

$$\tau = F.\ r.\ \sin(\theta)$$

Equation 1

Where τ is the torque, and it is measured in newton-meters (N. m), and F is the force required to make the turbine revolve, and it is measured in newtons (N), and r is the distance between the pivot point and the force, and it is measured in meters (M), and the is the angle between the force acting on the turbine and the distance between the pivot point and the force.

The torque is known by making the CFD on our 3D design and is determined by approximately **0.95** N . m, while the distance between the pivot point and the force, which is known when making the dimensions that meet the design requirements, and it is determined by **0.2** m, while θ is set to **90** to get the maximum force.

After knowing the torque and the distance between the pivot point and the force, and substituting for the 90-degree angle, all these given values were put in **Equation 1**, so the force was determined to be **4.75** N.

The rotation of the turbine is driven by the pressure difference between the inlets and outlets of each vent, as it produces a rotational force that acts perpendicular to the vertical axis of the turbine at a specific location. This quantity is the torque of the turbine (τ), which can be computed by **Equation 2:**

$$\tau = \blacktriangle P.\ r.\ A_{vent}$$

Equation 2

Where τ refers to torque, which equals **0.95** N · m, ΔP is the pressure difference, which equals **15** N, r is the distance between the vertical axis and the point where the force acts, which equals **0.2** M, and A denotes area.

After knowing the torque and the distance between the pivot point and the force, and the difference of pressure, all these given values were put in **Equation 2**, so the area was determined to be **0.316** m2.

Venturi effect

Venturi effect describes the fluid dynamics when the air flows through the inlet vents of the turbine, its velocity increases due to the small sectional area of the inlets, leading to a decrease in the pressure. The air exits the narrowed section and reaches the wide one; its velocity decreases, creating a high-pressure zone at the outlet. That pressure difference makes the airflow move from the entrance to the exit according to the pressure gradient, causing the turbine to revolve around a fixed axis.

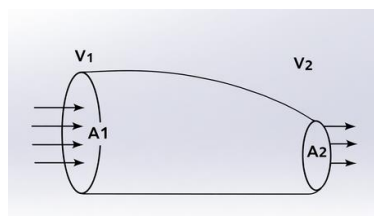

Figure 13: sectional area of external blade

The relation between the cross-sectional area of the inlets or outlets and the velocity is inversely proportional, **as shown in Figure 13**, and it's known as the equation of continuity. It describes the law of conservation of matter, where the volume flow rate through the streamline is constant (Serway & Vuille, 2016), **as shown in equation 3**. A is the cross-sectional area, and V is the velocity. The ratio between the entrance and the exit is **2:1**. Therefore, the velocity of the airflow at the entrance is **4.72** m/s, and the average velocity at the outlet is **5.8** m/s.

$$A1.\ V1 = A2.\ V2$$

Equation 3

The relation between the pressure and velocity is inversely proportional. It's described by Bernoulli's principle, which states that the sum of pressure, kinetic energy per unit volume, and the potential energy per unit volume has the same value at all points along a streamline (Serway & Vuille, 2016), **as shown in equation 4**. P is the pressure, ρ is the density of the fluid, V is the velocity of the fluid, g is the acceleration due to gravity, and h is the height above a set reference point. From equation 1. The velocity at the entrance is less than the velocity at the outlet, creating a high-pressure zone at the entrance with an average pressure of **101341** Pa and a low-pressure zone at the outlet with an average pressure of **101319.5** Pa. That pressure difference makes the turbine revolve.

$$P + \frac{1}{2}\rho V2 + \rho gh = \text{constant}$$

Equation 4

Angular momentum

In urban areas like Helwan, wind flow is turbulent, creating difficult conditions for wind turbines due to resistance from varying airflow directions. The effects of air resistance and changing wind directions are reflected in the turbine's angular momentum. Angular momentum describes the rotational motion retained in a system, even in the presence of air resistance. If the angular momentum changes, a torque (T) is

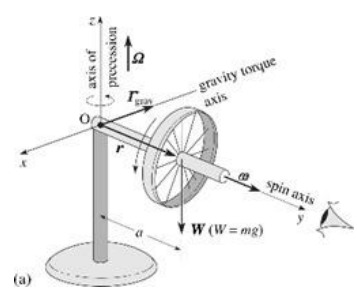

Figure 14: Angular momentum simulation

generated to maintain the rotational behavior of the system, **as shown in Figure 14**. Thus, increasing angular momentum results in a significant improvement in the durability of the prototype under harsh wind conditions. Angular momentum depends on two variables in **equation 5**

$$\boldsymbol{L = I\omega}$$

Equation 5

where I is the moment of inertia. For the o-wind turbine, which is considered a hollow spherical geometry, I is calculated by **equation 6**

$$\boldsymbol{I = \frac{1}{2}\, MR^2}$$

Equation 6

Where M is the mass of the sphere, and R is the radius. Furthermore, ω refers to the angular velocity, which represents the rotational speed of the turbine. Which is given by **equation 7**

$$\boldsymbol{\omega = \frac{2\pi \times RPM}{60}}$$

Equation 7

O-wind distributes mass uniformly at radius, which maximizes both I and angular momentum (L). This is particularly significant in Helwan's urban turbulence, where

wind direction and speed fluctuate continuously; however, this condition may cause the traditional turbines to stall repeatedly.

The calculation shows that the angular momentum for the prototype is approximately 40.4kg·m²/s. This angular momentum keeps the sphere rotating during wind fluctuations, thus contributing to a stable electrical output.

Power collection

The Betz Power Equation represents the power generated from the kinetic energy of wind, which is converted into electrical energy using wind turbines. This equation is fundamental in wind turbine design, as it ensures turbine efficiency. Therefore, it sets a theoretical limit for turbine performance and design. It is based on three main variables to calculate the output power. Firstly, air density (ρ) determines the mass of an exact volume of air. Secondly, the swept area (A) represents the area through which the wind passes, contributing to generating energy. Thirdly, wind speed (v) is the most critical factor, as it is cubed in **equation 8**.

$$\mathbf{P_w = \frac{1}{2}\rho A V^3}$$

Equation 8

In the tested location, it was approximately **5** m/s. Consequently, by integrating the principle of wind power with the O wind prototype, the required electrical output was successfully achieved, generating approximately **0.98W** within five minutes.

This value represents the theoretical energy of the prototype. However, the actual electrical energy is calculated using **equation 9**

$$\mathbf{P = IV}$$

Equation 9

where I is the current, V is the voltage. The resulting power is approximately **0.5W.** To evaluate the efficiency of the prototype, the power coefficient (C_p) is used, which is the ratio of the mechanical power by the turbine to the theoretical power available in the wind. Using **equation 10**, the calculated power coefficient is approximately **51%**, thus contributing to the evaluation of the prototype's performance. Furthermore, it allows comparing it with other turbine designs, where the prototype C_p effectively reflects its efficiency, ultimately giving feedback on the prototype.

$$c_P = \frac{P_{eff}}{p_{wind}}$$

Equation 10

Table 2: Learning outcomes

Subject	LO	Concept	Connection
Physics	PH.1.09	Bernoulli equation	To determine the relation between the pressure and the velocity at the inlet and the outlet.
Physics	PH.1.08	Pressure difference and force	To adjust the inlets and the outlets according to the pressure gradient to make the turbine rotate.
Physics	PH.1.05	Torque	To know the force that the turbines need to revolve.
Geology	ES.1.10	Wind energy through turbines	to gain more knowledge about wind energy through turbines.

Conclusions

Egypt stand up to different grand challenges like the science and technological weaknesses and the lack of alternative energy. As a result, the reliance on fossil fuels and an increase in greenhouse gases. To solve that, the project approach adopted eco-friendly materials with at least one recycled sustainable material, which makes the prototype greener and environmentally friendly. Additionally, the prototype is designed as an O-type, that is, a spherical anthropomorphic wind turbine with a tetrahedron shape to capture turbulent wind omnidirectionally, for capturing from vertical, horizontal, and diagonal directions. The prototype has enhanced its output by **10%,** therefore producing **170** joules in **5** minutes. Additionally, the volume doesn't exceed **0.25** by making the prototype **0.128** cubic meters; thus, the project meets the design requirements effectively and successfully.

Unlike the Zafarana wind station, where maintenance challenges arise due to harsh conditions, however, issue is minimized in prototype, as it is designed for an urban location such as Helwan, which is an urban location. Moreover, unlike the Ras Ghareb wind farm, which has a high cost due to technology requirements, on the other hand, the prototype has a low cost comparatively. Consequently, the prototype successfully demonstrated its efficacy and sustainability as a solution for generating electricity instead of fossil fuels.

Recommendation

Real-life application

The project is recommended to be located at Helwan, **as shown in Figure 15**, which is located in the southeastern part of Cairo, about **25** kilometers from the city center. It faces a significant problem, namely, the usage of harmful fossil fuels in electricity generation plants.

Figure 15: Location of Helwan

The location suffers from the problem, as it contains one of the largest power plants in Egypt, the South Helwan power plant. Moreover, it's an oil-fueled steam power plant using natural gas as primary fuel and diesel (mazut) as an alternative

fuel. It's located on the eastern bank of the Nile River. **15** km north of Beni Suef city. Generating **1.95** GW, equivalent to about **90%** of the output of the Aswan High Dam. The efficiency of the proposed steam power plant is **42-45%** with natural gas, with associated CO2 emissions of about **520** g/kWh. Electricity generation contributes to environmental pollution and affects residents. Due to high PM **2.5** Levels in the area. Helwan suffers from severe air pollution. That reaches **35** $\mu g/m^3$ at its peak, which is higher than the safe limit, which is **5** $\mu g/m^3$. This is what the project aims to achieve.

The real-life dimensions of the project will be **2.2** m in length and **2.2** m in width. The turbine's pipe will be made of steel due to its hardness, strength, and great durability. The wind turbine itself will be made of aluminum. Due to its low weight, which makes it easy to revolve. The base of the turbine will be made out of concrete to support the weight of the turbine.

Carbon fiber composites

Carbon fiber composites (HMCF UD) are recommended to be used in the turbine from the outside of the turbine, **as shown in Figure 4.1.4**. It has a high Young's modulus with a value of **175** GPa, which is higher than that of aluminum. Providing more resistance under intense air pressure and a longer lifespan. and a low-density of **1600** Kg/M3. Which is lower than that of aluminum, with a percentage of **40.74%.** resulting in lighter structural weight. That provides the turbine with a very low inertia, making the turbine start up at low wind speeds. and less load on the mechanical parts. It can be shaped into thinner and smoother, resulting in better airflow interactions. It wasn't used due to its high-cost of **1862** EGP per **1** M2. In addition, no local availability in Egypt.

Axial-Flux Permanent Magnet Generators (AFPM)

The Axial-Flux Permanent Magnet Generator is recommended to be used instead of an RS-545SH-2190 DC motor, **as shown in Figure 4.1.5**, as it is built with a high-power ratio. Where the generator generates large power despite its small size compared to the DC motor used, it has a high torque and low rpm, meaning it can operate easily in poor air conditions and is suitable for cities' turbulent wind conditions, similar to what we need. The design is characterized by its lightweight and low noise and vibration levels due to its low-friction flow. The generator wasn't

used due to its high cost and complexity, in addition that it is not locally available in Egypt.

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